

object is read by Arduino Nano in centimetres (cm) and displayed on the GLCD screen. If the distance is more than 99cm, the display shows 'OUT' to indicate that the object is out of range. The object detected by the lidar at 40cm distance, along with the light rays, is shown in Fig. 3.

There are four LED indicators in the circuit. LED1 is power indicator while LED2 (OBST) indicates that there is an object within the 99cm range at every movement. LED3 (LEFT) and LED4 (RIGHT) indicate whether the object is on left side or right side of the area being scanned. If the object is in the middle, both LED3 and LED4 glow.

Following points regarding this system are worth noting:

1. The lidar may not detect transparent obstacles as it works on reflection of light.
2. To change the direction of stepper motor rotation, just reverse the connections of stepper motor to A4988 driver.
3. The current limiting of stepper motor may be controlled by adjusting the preset on A4988 module (refer A4988 module data sheet), if required.
4. The following variables may be modified in the sketch (then

uploaded) to match individual need, if any:

```
const int STEPPER_DELAY = 100; // delay
to settle the stepper
const int stepsPerRevolution = 200; //
set the number of steps per revolution
for your motor
int ONESTEP=2;//number of step signals
sent to A4988 per loop
```

Before compiling and uploading the code to Arduino Nano, you need following files, which can be downloaded from <https://github.com/pololu/vl53l0x-arduino>:

VL53L0X.cpp
VL53L0X.h
GLCDfont.h
license.txt

This circuit is useful for an obstacle-detection robotics project to know the distance of the obstacle. **EFY**



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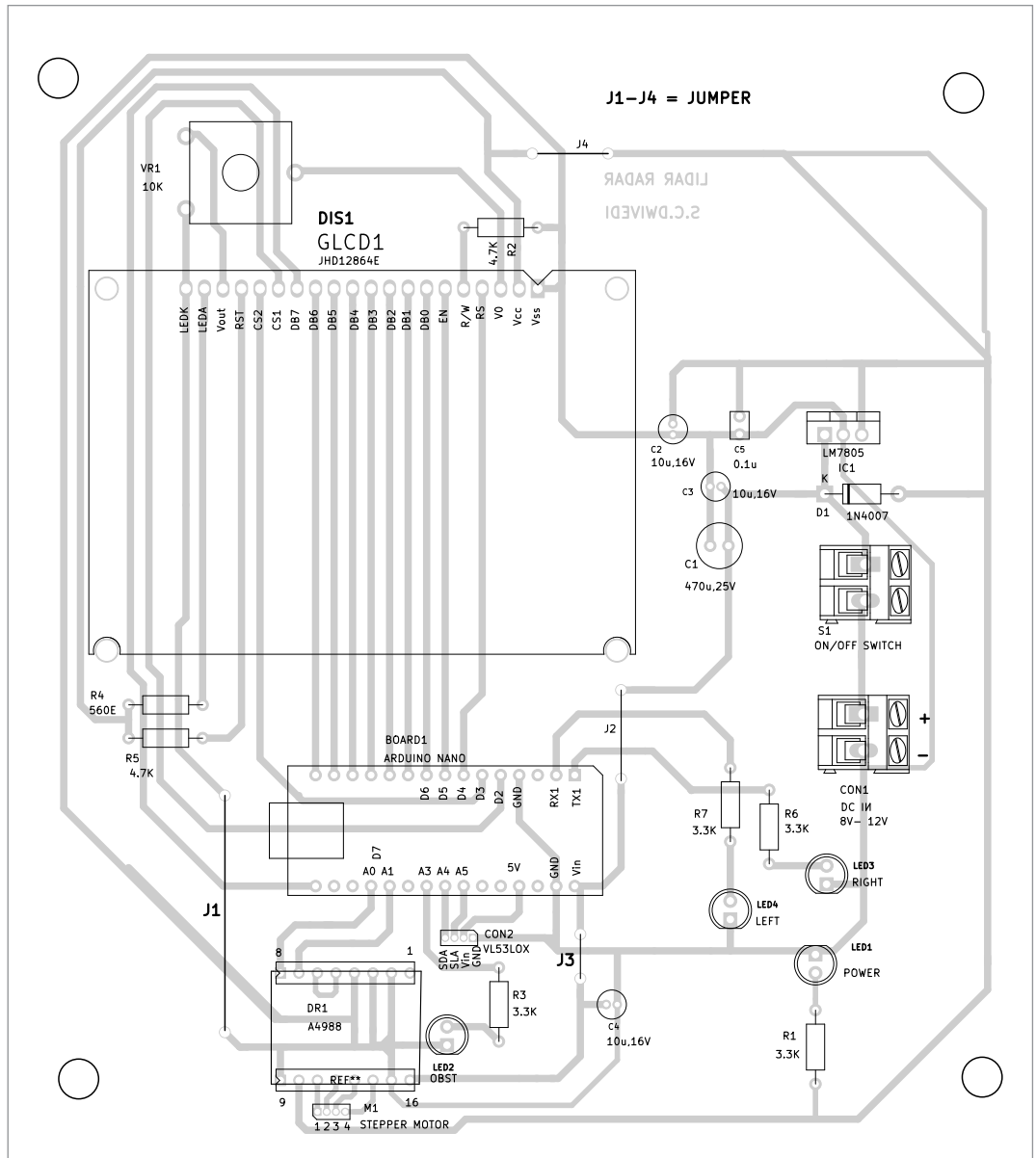


Fig. 5: Component layout of the PCB